

1, Describe how an incremental encoder detects the direction of rotation. An absolute encoder has to measure angles of rotation as small as 2° . How many tracks should it have?

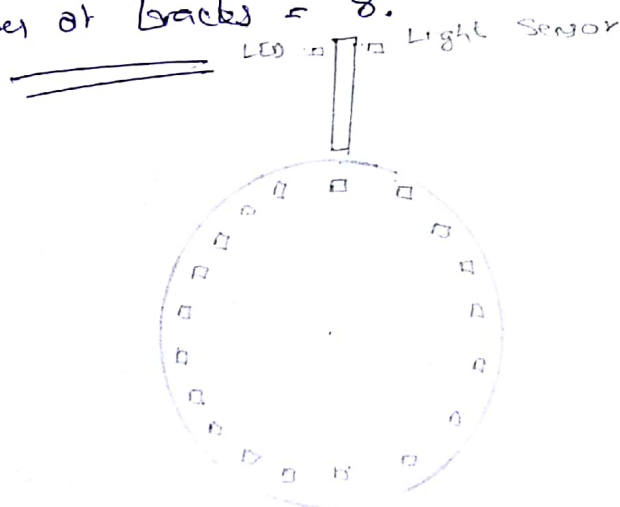
Ans. A beam of light passes through slots in a disc and is detected by a suitable light sensor. When the disc is rotating, a pulsed output is produced by the sensor with the number of pulses being proportional to the angle through which the disc rotates. Thus the angular position of the disc and hence the shaft rotating it can be determined by the number of pulses produced since some datum position.

Given: For the absolute encoder the resolution is 2°

$$\text{ie } \frac{360}{\text{no. of position}} = 2^\circ \Rightarrow \text{no. of position} = 180$$

$$2^{\text{no. of tracks}} \geq 180$$

$\Rightarrow$ number of tracks = 8.



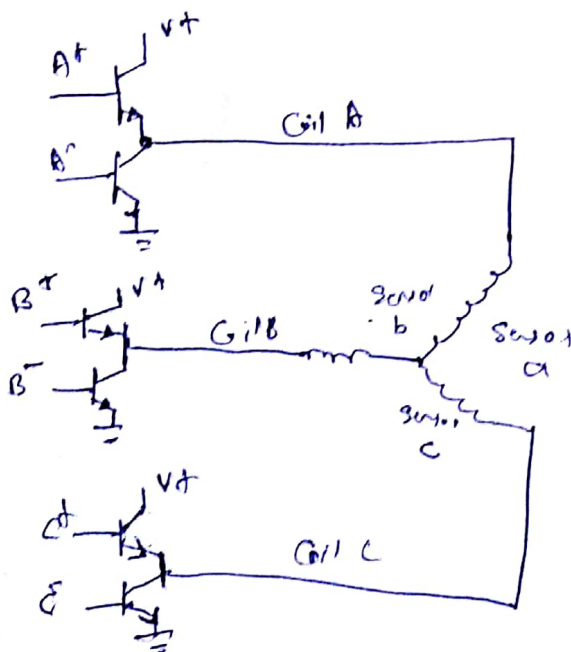
2. Describe bonded strain gauge. Explain how it can be used to measure torque.

Bonded strain gauges are directly bonded, i.e. pasted on the surface of the structure under study. Hence they are termed as bonded strain gauges. The three types of bonded strain gauges are

- * Wire strain gauge
- * Metal foil strain gauge
- * Semi-conductor gauge.

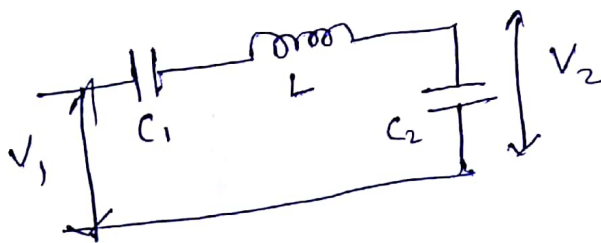
Strain gauge converts torque into a change in electrical resistance. The strain gauge is bonded to a beam or structural member that deforms when a torque or force is applied. Deflection induces a stress that changes its resistance. A Wheatstone bridge converts the resistance change into a calibrated output signal.

3. Explain transistor switching in a permanent magnet DC motor.



To switch the coils in sequence we need to supply signals to switch the transducers on in the right sequence. This is provided by the output from the three sensors operating through a decoder circuit to give the appropriate base currents. They when the rotor is in the vertical position, i.e. 0° , there is an output from the sensor C but none from A and B. This is used to switch on transistor A^+ and B^- . For the rotor in the 60° position there are signals from the sensors B and C and therefore A^+ and C^- are switched on.

A switching circuit is used to convert one level of dc voltage to an output dc voltage. The filter circuit to filter out the high frequency is shown in the figure. Calculate the transfer function $V_2(s)/V_1(s)$.



$$V_1(s) = L \frac{di}{dt} + \frac{1}{C_1} \int i dt + \frac{1}{C_2} \int i dt$$

$$V_2(s) = \frac{1}{C_2} \int i dt$$

Taking Laplace transform

$$V_1(s) = Ls I(s) + \frac{1}{C_1 s} I(s) + \frac{1}{C_2 s} I(s) = \left[Ls + \frac{C_2 + C_1}{C_1 C_2 s} \right] I(s)$$

$$V_2(s) = \frac{1}{C_2 s} I(s)$$

$$\frac{V_2(s)}{V_1(s)} = \frac{\frac{1}{C_2 s}}{\left[Ls + \frac{C_2 + C_1}{C_1 C_2 s} \right]} = \frac{\frac{1}{C_2}}{Ls^2 + \frac{C_2 + C_1}{C_1 C_2}}$$

$$= \frac{1}{C_2} \cdot \frac{C_1 C_2}{Ls^2 C_1 C_2 + C_2 + C_1} = \frac{C_1}{Ls^2 + C_2 + C_1}$$

7. The rotational velocity of the satellite shown in figure adjusted by changing the length of beam L . The transfer function between $\omega(s)$ and the incremental change in beam length $\Delta L(s)$ is $\frac{\omega(s)}{\Delta L(s)} = \frac{5(s+2)}{2(s+5)(s+1)^2}$. The beam length

is $\Delta L(s) = \frac{1}{4s}$. Determine the response of the velocity $\omega(s)$

$$\Delta L(s) = \frac{1}{4s}$$

$$\frac{\omega(s)}{\frac{1}{4s}} = \frac{5(s+2)}{2(s+5)(s+1)^2}$$

$$\omega(s) = \frac{5(s+2)}{8s(s+5)(s+1)^2}$$

$$\frac{5}{8} \left[\frac{A}{s} + \frac{B}{s+5} + \frac{C}{s+1} + \frac{D}{(s+1)^2} \right]$$

$$\frac{5}{8}(s+2) = A(s+5)(s+1)^2 + B s(s+1)^2 + C(s+1)(s+5)s + D(s+5)s$$

$$A = \frac{1}{4}, \quad B = \frac{3}{128}, \quad D = -\frac{5}{32}, \quad C = -\frac{35}{128}$$

$$\mathcal{L}^{-1} \omega(s) = \mathcal{L}^{-1} \frac{1}{4s} + \mathcal{L}^{-1} \frac{3}{128(s+5)} + \mathcal{L}^{-1} \frac{-35}{128(s+1)} - \mathcal{L}^{-1} \frac{5}{32(s+1)^2}$$

$$\omega(s) = \frac{1}{4} + \frac{3}{128} e^{-5t} + \frac{-35}{128} e^{-t} - \frac{5}{32} e^{-t} t$$

9. A room at temperature 'T' is cooled by an ac, which removes heat from the room at rate of Q_0 from surrounds, which is at a temperature T_0 , heat flows into the room across a thermal resistance of R. The thermal capacitance of the room is C. Model the system.

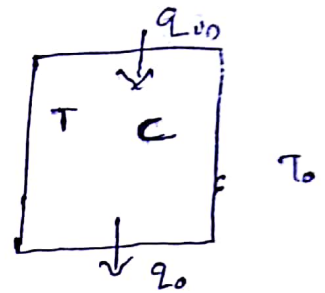
the heat into the room is Q_{in} .

~~The heat~~

Since the room has thermal capacity C,

$$Q_{in} - Q_0 = C \frac{dT}{dt} \quad \text{--- (1)}$$

$$Q_{in} = \frac{T_0 - T}{R} \quad \text{--- (2)}$$



where R is the resistance of the walls

$$\frac{T_0 - T}{R} - Q_0 = C \frac{dT}{dt}$$

$$\Rightarrow \underline{\underline{RC \frac{dT}{dt} + T = T_0 + RQ_0}}$$

10. The door of a public transport vehicle is opened and closed by a double acting pneumatic cylinder. Draw the hydraulic and electric circuit diagrams, showing suitable solenoid-controlled direction control valves and switches for opening and closing.

